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Clockwise tectonic rotation of Tertiary sedimentary basins in central Hokkaido, northern Japan

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ABSTRACT

Paleomagnetic data from Tertiary sedimentary strata in the Yubari and Haboro areas of central Hokkaido indicate consistently clockwise-deflected magnetic declinations. The mean declination of middle middle Miocene rocks deviates 21° ±15° from the Eurasian reference direction, and that of the Oligocene to lower middle Miocene rocks deviates 30° ±21°. This suggests that these sedimentary basins have rotated 10°–30° clockwise about a vertical axis since the middle Miocene. These clockwise rotations can be explained by a right-lateral displacement of a shear zone in central Hokkaido. This study demonstrates that central Hokkaido is not simply a single tectonic unit of northeast Japan characterized by counterclockwise rotations from early to middle Miocene time.

INTRODUCTION

Structural analyses (e.g., Kimura et al., 1983) have indicated that central Hokkaido is a zone of deformation characterized by north-northwest–trending folds and faults (Fig. 1). Jolivet and Huchon (1989) have interpreted this zone as a right-lateral, transpressional shear zone formed during Oligocene to middle Miocene time. Recently, the central Hokkaido shear zone has been the focus of much attention in terms of the formation of the Japan Sea (e.g., Jolivet et al., 1991). We conducted a paleomagnetic study on Tertiary sedimentary rocks exposed in the Haboro and Yubari areas, which are 150 km apart within this shear zone (Fig. 1), to document the tectonic deformation of these sedimentary basins and to provide paleomagnetic delimitations on the movement of this shear zone.

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GEOLOGIC SETTING

Hokkaido is composed of three major geologic provinces: west, central, and east Hokkaido (Fig. 1). West Hokkaido consists of Mesozoic to Cenozoic sedimentary and volcanic complexes with abundant granite intrusions. Central Hokkaido is underlain by Tertiary to Cretaceous clastic deposits and highly deformed Cretaceous sedimentary rocks accompanied by Tertiary plutonic and metamorphic rocks. East Hokkaido consists mainly of sedimentary and volcanic rocks of Late Cretaceous to early Tertiary age.

Tertiary clastic deposits are well exposed in western central Hokkaido. The Neogene, mostly Miocene, sequences are unconformably underlain by Paleogene coal-bearing sandstone and siltstone. The Miocene de-

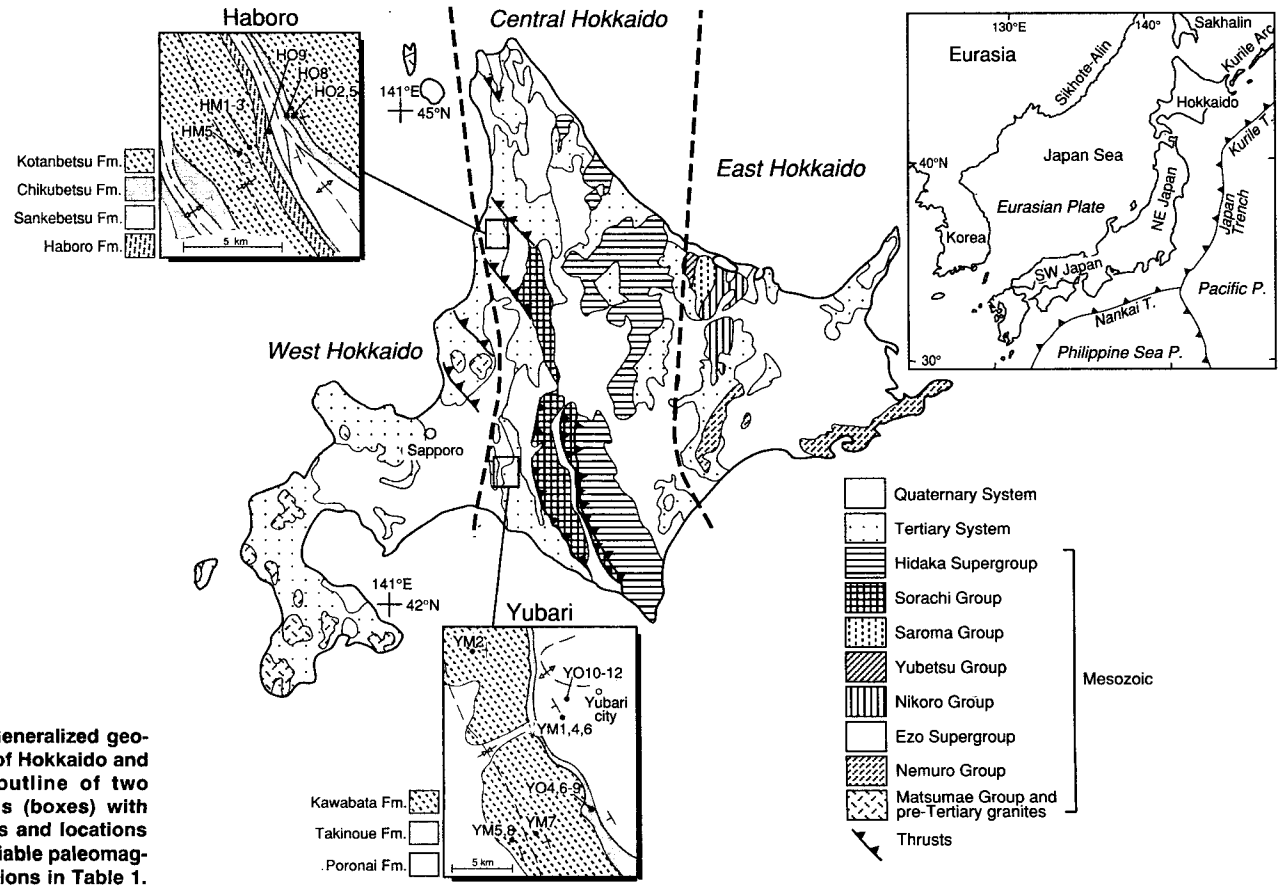


Figure 1. Generalized geologic map of Hokkaido and geologic outline of two study areas (boxes) with site indexes and locations yielding reliable paleomagnetic directions in Table 1.

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posits are principally turbidites derived from sources to the east by the uplift of the central Hidaka belt (Miyasaka et al., 1986). These strata are folded, with associated secondary thrusts that display northwest-trending en echelon structures typical of a right-lateral strike-slip zone (Kimura et al., 1983).

Two datum horizons of calcareous nannoplankton, identifiable as CP14 (40–45 Ma) and CP16 (35–37 Ma), were recognized in a Paleogene formation (Sankebetsu Formation) of the Haboro area (Okada, 1982). Planktonic foraminifera and pollen indicate a late Eocene or late Eocene to early Oligocene age for the lower part of this formation (Maiya et al., 1982). The upper part has been dated as late Oligocene to early Miocene by diatom and silicoflagellate data (Kanno and Matsuno, 1960), and a K-Ar age of 19.3 ±1.0 Ma was reported for an acidic tuff in the uppermost horizons (Hoyanagi and Matsui, 1985). A planktonic foraminiferal biostratigraphic study yielded late Eocene to early Oligocene ages for the Poronai Formation in the Yubari area (Kaiho, 1984). The Takinoue Formation in Yubari and the Chikubetsu Formation in Haboro commonly yield marine mollusks found widely in the lower middle Miocene deposits in the Japanese islands (Chinzei, 1986). Two overlying formations, the Kawabata Formation in Yubari and the Kotanbetsu Formation in Haboro, correlate to a diatom zone, the *Denticulopsis lauta* zone, indicating dates between about 13 and 15.5 Ma (Koizumi, 1985). This biostratigraphic and chronostratigraphic information is summarized in Figure 2.

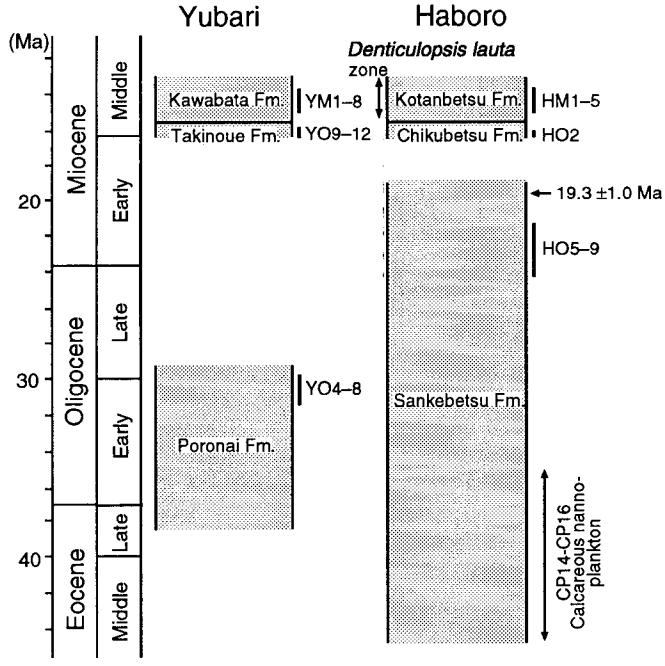


Figure 2. Stratigraphy of Tertiary series in Yubari and Haboro areas, central Hokkaido, showing horizons of reliable paleomagnetic sites in Table 1 (vertical bars) and biostratigraphic age range and radiometric age (arrows).

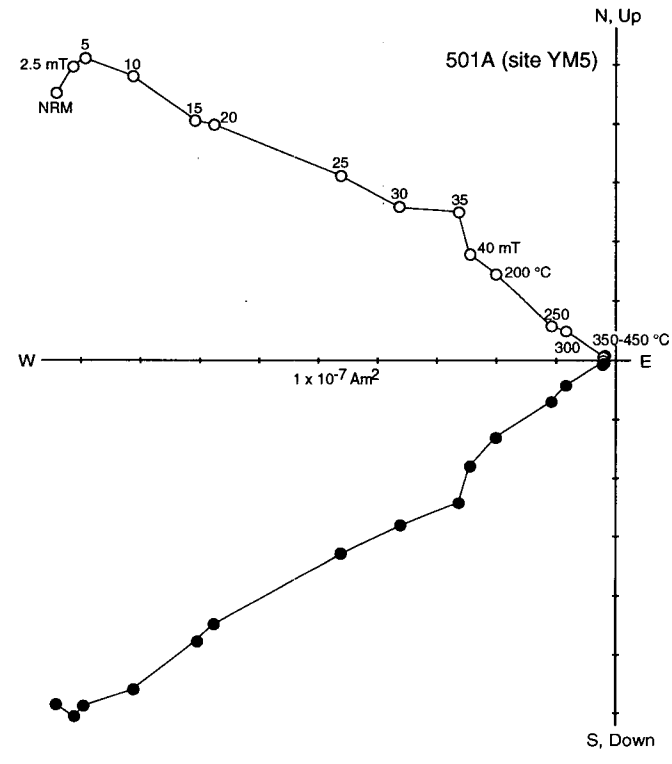


Figure 3. Example of orthogonal vector diagrams of combined alternating-field and thermal demagnetizations. Solid circles are projections on horizontal plane, and open circles are projections on east-west vertical plane, in stratigraphic coordinates. Unit is bulk magnetic moment. Site YM5 in Kawabata Formation, Yubari area. Reversed polarity with characteristic direction, declination = 208°, inclination = −32°, calculated by least-squares fit to steps above 200 °C.

PALEOMAGNETIC METHODS AND RESULTS

We collected samples from 21 sites in the Oligocene to lower Miocene formations and from 38 sites in the middle Miocene formations. To perform fold tests, site locations were distributed evenly among different fold limbs. A total of 817 specimens were measured with a three-axis cryogenic magnetometer. All the specimens were subjected to progressive thermal or alternating-field demagnetization according to magnetic coercivity. Characteristic magnetization components were isolated by applying principal-component analysis (Kirschvink, 1980). All the vector segments that could be defined by at least four endpoints with a MAD (maximum angular deviation) of <15° underwent further data analyses. Site-mean directions with α_{95} values of <20° were judged to be acceptable. An example of the progressive demagnetization is shown in Figure 3.

A total of 15 sites in Yubari and eight sites in Haboro provided acceptable mean directions. The results are summarized in Table 1, which is subdivided into two groups: the middle middle Miocene and the Oligocene to early middle Miocene, because it has been suspected that the present framework of the Japanese arc was formed between these two age intervals (e.g., Otofuji et al., 1991). The site-mean directions before and after bedding-tilt correction are shown in Figure 4. The unfolded directions are divided into two tightly grouped, antipodal assemblages. The antipodal relation is significant on the basis of statistical tests (McFadden and McElhinny, 1990), except for the Oligocene–early middle Miocene Yubari directions, which lack normal polarity. The change in grouping was proven statistically significant by the method of McFadden and Jones (1981), except for the middle middle Miocene Haboro data.

We obtained a reference direction for the middle middle Miocene from the Neogene mean pole (lat 88°N, long 147°E, A_{95} = 7°) that was defined by using mean poles of Bretstein (1988) for Shikhote-Alin and those of Zhao et al. (1990) and Zheng et al. (1991) for

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TABLE 1. PALEOMAGNETIC SUMMARY OF TERTIARY SEDIMENTARY ROCKS IN CENTRAL HOKKAIDO, NORTHERN JAPAN

Site	Location	N / n	Str / Dip	D_c / I_c	D / I	α_{95} / κ	VGP	F / ΔF	R / ΔR
Yubari (middle middle Miocene)									
YM1	42.94/141.98	6/5	150/42	343/+72	37/+41	13/39	72/111		
YM2	43.10/141.84	11/7	180/48	204/–45	228/–15	9/26	65/265		
YM4	42.94/141.98	7/6	146/52	49/+53	52/+01	13/24	51/230		
YM5	42.95/141.88	8/6	338/30	201/–25	184/–42	8/55	55/285		
YM6	42.94/141.97	11/10	145/60	199/–47	211/+06	11/19	69/270		
YM7	42.95/141.88	6/6	152/73	212/–48	221/+20	10/30	61/250		
YM8	42.95/141.88	9/7	337/28	169/–35	148/–36	8/44	64/347		
Mean				18/+48	29/+17	16/16 (5)	71/266	–16/13	18/20
							A_{95} = 18		
Yubari (Oligocene–early middle Miocene)									
YO4	42.98/141.98	6/6	148/26	195/–40	204/–19	10/56	66/287		
YO6	42.98/141.97	6/6	140/26	185/–58	202/–37	14/26	85/276		
YO7	42.98/141.97	7/7	140/26	210/–64	219/–38	17/17	69/216		
YO8	42.98/141.98	8/7	132/20	181/–62	196/–45	13/21	89/223		
YO9	42.97/141.96	6/5	155/30	254/–60	250/–31	15/27	37/207		
YO10	42.95/141.99	10/8	145/78	233/–57	234/+21	12/19	50/221		
YO11	42.95/141.98	11/6	140/70	220/–63	225/+07	5/98	61/215		
YO12	42.95/141.99	11/6	140/65	98/–72	214/–36	16/9	39/99		
Mean				205/–64	219/–24	14/16 (8)	72/208	–3/12	26/29
							A_{95} = 22		
Haboro (middle middle Miocene)									
HM1	44.37/141.88	10/10	152/50	197/–57	219/–14	7/57	75/255		
HM2	44.37/141.88	8/7	152/50	226/–70	236/–21	6/91	59/198		
HM3	44.37/141.88	7/5	152/50	220/–53	229/–05	7/83	57/237		
HM5	44.37/141.87	6/4	162/25	19/+36	29/+19	18/15	61/283		
Mean				28/+54	43/+15	19/25 (36)	67/241	–10/15	28/27
							A_{95} = 21		
Haboro (Oligocene–early middle Miocene)									
HO2	44.40/141.91	8/8	311/45	26/+72	232/+63	10/30	70/187		
HO5	44.40/141.91	9/8	301/45	53/+74	200/+59	8/39	55/185		
HO8	44.38/141.88	10/8	160/65	38/+56	53/–05	7/50	60/234		
HO9	44.38/141.88	9/9	160/60	214/–52	229/+02	7/51	61/244		
Mean				37/+64	65/+55	13/48 (2)	64/213	–4/11	38/27
							A_{95} = 18		
Overall mean (middle middle Miocene)									
				22/+51	35/+17	11/19 (8)	70/255	–13/9	21/15
							A_{95} = 12		
Overall mean (Oligocene–early middle Miocene)									
				29/+64	42/+35	10/22 (4)	70/210	–3/9	30/21
							A_{95} = 14		

Note: Location = latitude / east longitude; N / n = number of specimens used in statistics / number of independently oriented cores; Str / Dip = strike / dip (clockwise from strike); D_c / I_c = tilt-corrected declination / inclination; D / I = in situ declination / inclination; α_{95} / κ = radius of 95% confidence circle / Fisher's precision parameter. The statistical parameters were calculated for unfolded directions; κ for in situ directions are in parentheses; VGP = latitude / east longitude of virtual geomagnetic pole; and A_{95} = 95% confidence limit for mean VGP. F / ΔF, R / ΔR are flattening and rotation indices of Beck (1980) with errors on 95% confidence level corrected using factors of Demarest (1983). Reference directions (Decl. / Incl.): 0/64 (middle middle Miocene) and 359/67 (Oligocene-early middle Miocene) for the Yubari area at 43°N, 142°E, and 0/65 (middle middle Miocene) and 359/68 (Oligocene-early middle Miocene) for the Haboro area at 44°N, 142°E.

northern China. Because of the lack of reliable Paleogene data from eastern Asia, the Oligocene–early middle Miocene reference was calculated by using the Oligocene pole of Europe (83°N, 141°E, A_{95} = 6°; Westphal et al., 1986). The differences between our data and these reference directions are summarized in Table 1 and indicate consistently clockwise declination shifts. The average middle middle Miocene declination for both areas deviates eastward from the reference by 21° ±15°, and the Oligocene–early middle Miocene declination deviates by 30° ±21°.

DISCUSSION

Paleomagnetic data from Hokkaido are sparse compared to elsewhere in Japan. Preliminary studies of the Tertiary sedimentary and volcanic rocks in the Oshima Peninsula of west Hokkaido indicated a post-Oligocene counterclockwise rotation (Fujiwara and Sugiyama, 1986). A similar counterclockwise rotation was reported for Cretaceous granites in this area (Ito and Tokieda, 1986). These rotations are comparable with previous results from northeast Japan (e.g., Tosha and Hamano, 1988). Central Hokkaido is unique in its clockwise rotation in comparison to west Hokkaido.

Rotations about a vertical axis are commonly observed in areas crosscut by strike-slip faults. For instance, terranes in the western Transverse Ranges in California bounded by right-lateral faults have rotated clockwise by several tens of degrees (e.g., Hornafius et al., 1986). Many fault-bounded blocks in the Basin and Range province of continental extension underwent clockwise or counterclockwise rotations of various amounts (e.g., Janecke et al., 1991). Accordingly, the clockwise rotation in central Hokkaido may be explained by north-trending right-lateral shear deformation. However, unlike these other areas, the sedimentary basins are not crosscut by strike-slip faults but are accompanied by en echelon folds and thrusts. These structures suggest that the rotations cannot simply be modeled by vertical-axis rotations of rigid, crustal blocks. It may be more realistic to infer that the rotations resulted from ductile deformation of blocks with internal folding (Fig. 5).

Lallemand and Jolivet (1986) interpreted the Japan Sea to be a right-lateral pull-apart basin that opened along the central Hokkaido shear zone. Jolivet et al. (1991) concluded that the pull-apart opening had ended by the end of middle Miocene time, as did the shear motion in central Hokkaido. Our paleomagnetic data from central Hokkaido, however, indicate that the shear motion responsible for the clockwise rotation persisted after middle Miocene time. In addition to our data, preliminary data from upper Miocene rocks in the Haboro area (T. Takeuchi, unpublished) also suggest a clockwise rotation of a magnitude comparable to that of the middle Miocene rotation. Thus, the dextral shear may have been active until post-Miocene time, and most of the clockwise rotations may have taken place after the late Miocene. Ocean Drilling Project results (Tamaki et al., 1990) indicate that the Japan Sea opened during early to middle Miocene time; thus, these post-Miocene rotations cannot be related to the back-arc spreading. This conclusion suggests that the shear zone may never have played an important role in

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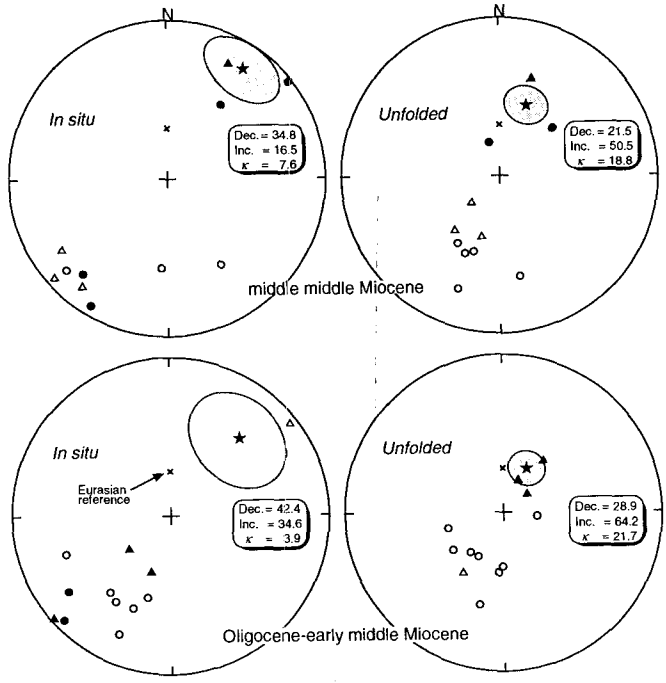


Figure 4. Mean directions at each site of middle middle Miocene (top) and Oligocene to early middle Miocene (bottom), in geographic (left) and stratigraphic (right) coordinates. Circles are directions from Yubari area, and triangles are directions from Haboro area. Solid symbols are equal-area projections on lower hemisphere, and open symbols are equal-area projections on upper hemisphere. Stars and shaded ovals indicate overall mean directions (Dec. = declination, Inc. = inclination) and their associated errors at 95% confidence level (κ), which were calculated by converting reversed directions to normal. Xs are reference directions of Eurasia.

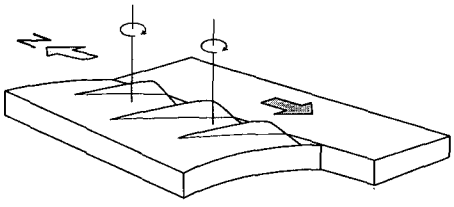


Figure 5. Model for tectonic rotation of Tertiary sedimentary basins in central Hokkaido. Drag along right-lateral shear zone produces clockwise rotation about vertical axis as well as en echelon folds and thrusts.

the opening of the Japan Sea; the eastern boundary of the pull-apart opening may not be the central Hokkaido shear zone but should be located more to the west than previously predicted. This study demonstrates that the popular hypothesis of the pull-apart opening model of the Japan Sea requires some substantial refinement.

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